

BUILD YOUR OWN M2 NODE

What it costs. What it takes. What you get.

CCC26
LFHI



RECOMMENDED

~\$1,385

Dual Raspberry Pi 5 16GB
Full redundancy. Both service stacks.
Best for active chapters with regular deployments.

BUDGET BUILD

~\$889

Single Raspberry Pi 5 16GB
Comms stack only or tactical stack only.
Good starting point. Expand later.

WHAT YOU SAVE

- Monthly subscription: \$0
- Platform dependency: none
- Data handed to third parties: none
- Permission required to operate: none

BILL OF MATERIALS — RECOMMENDED BUILD

Component	Qty	Approx. Cost	Purpose
Raspberry Pi 5 16GB	2	\$160 ea	Compute nodes (comms + tactical)
Pi 5 active cooler	2	\$10 ea	Thermal management
Samsung 256GB microSD	2	\$28 ea	OS + boot media
Samsung 2TB NVMe SSD	1	\$120	Monero blockchain + data persistence
NVMe HAT for Pi 5	1	\$25	NVMe interface
8U 10" mini rack	1	\$85	Enclosure
1U 10" rack shelf	2	\$18 ea	Pi mounting
GL.iNet Slate AX router	1	\$80	WiFi gateway + LAN
TP-Link 8-port switch	1	\$18	Managed LAN
Heltec LoRa V3	2	\$25 ea	LoRa radios (Reticulum + Meshtastic)
Tripp Lite BC600R UPS	1	\$55	Power continuity
Anker 747 GaN 150W	1	\$50	Multi-port charging
Tupavco 8-outlet PDU	1	\$30	Rack power distribution
7" touchscreen	1	\$75	Kiosk display
Cables, hardware, misc	—	~\$50	USB, Ethernet, SMA, etc.
TOTAL		~\$1,385	

HOW LONG DOES IT TAKE?

Weekend 1 Hardware assembly, OS installation, basic networking

Weekend 2 Docker services, Matrix, ATAK, LoRa configuration

Weekend 3 Hardening, field test, documentation, field card printing

Future events < 1 hour deploy time once built and tested

WHAT SKILLS DO YOU NEED?

- ✓ Linux command line (SSH, apt, systemctl)
- ✓ Basic networking (IP addresses, ports, DNS)
- ✓ Patience and a build guide

The rest is documented.

13 phases. Every command included.

Start building.

Full build guide, BOM with Amazon links, and all config files:

github.com/PapaSierra555/M2-Community-Node

Scan QR → open project, clone repo, or read docs online.



Scan to open